Tell the computer

Optimization · Level 2 · Lesson 4



In this lesson you will

- Read a small model in Python: variables, limits, a goal.
- Put a model’s lines in order and fill its gaps.
- Predict what it prints, and run it in the lab.

Key ideas

- Model, variables, limits, goal, solve
- <= at most, >= at least
- OPTIMAL and INFEASIBLE
- The solver knows only your rules

Activity: Predict, then run

1. Read (5 min): Find the variables, limit and goal.
2. Predict (5 min): Write what three changes will do.
3. Run (7 min): Run each change and compare.
4. Share (3 min): Which prediction was wrong, and why?

Python downloads the first time (about 0.3 MB): run one example while online. No phone? Do the activity on paper with the garden model.

Let’s discuss

1. Which line of the program is the goal, and how can you tell?
2. When would you get INFEASIBLE?
3. What rule would make a school plan fairer?

Exit ticket

- Write one limit as a model.add(...) line and say what it means.